

Event & Room Occupancy Management System

Project Review 2 — Design, Progress & Demonstration

TEAM

Sec 10 · Team 1

TRIMESTER

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MEMBERS

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Problem Statement

Students and administrators lack a single, reliable way to see which campus rooms are free and to manage events without last-minute chaos.



Room availability is invisible

Students struggle to find free rooms during college hours — there is no shared, real-time view.



Spot registrations disrupt planning

Hackathons, workshops and seminars often see sudden participant surges beyond initial estimates.



Conflicts go undetected

Manual coordination leads to double-booked rooms and scheduling clashes.



Space is used inefficiently

Without occupancy tracking, capacity mismatches and overcrowding are common.

Objectives

- 1 Provide a centralized platform for room and event management across campus
- 2 Give students real-time visibility into room availability and schedules
- 3 Streamline event creation, room allocation and participant registration
- 4 Detect and prevent room-booking conflicts before they occur
- 5 Track expected vs. actual occupancy to support data-driven decisions
- 6 Improve overall utilization of campus spaces and reduce manual effort

Proposed System

A centralized, web-based platform where students discover rooms and events, and administrators allocate space, monitor registrations, and track real occupancy — all from one dashboard.



Live Room Availability

View real-time room status, capacity and schedules before booking.



Event Scheduling

Admins create and manage events, allocate rooms and set expected turnout.



Participant Registration

Students register for events; the system tracks registrations against capacity.



Conflict Detection

Automatically flags overlapping bookings and capacity mismatches.



Admin Dashboard

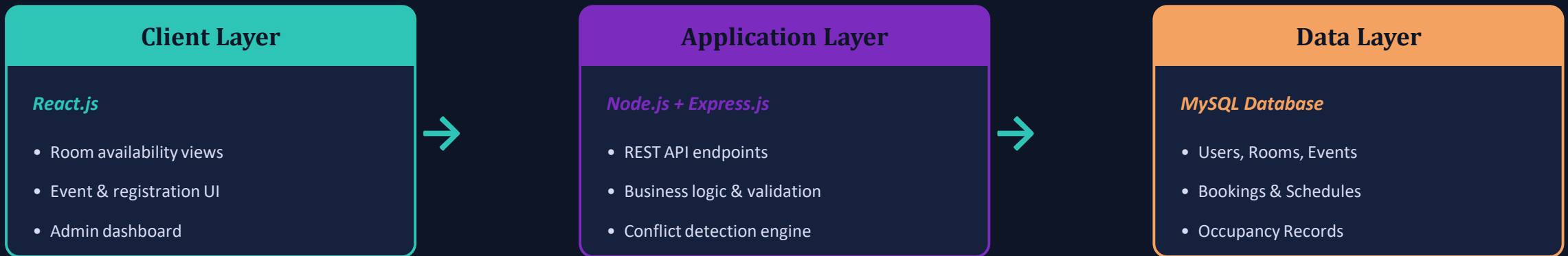
A single console to monitor bookings, rooms and occupancy trends.



Occupancy Tracking

Compares expected vs. actual attendance to guide future planning.

System Architecture



Client requests flow through the Express REST API, which applies business rules (e.g. conflict checks) before reading or writing to MySQL — keeping presentation, logic and data cleanly separated.

Modules / Functionalities



User Management

Registration, login and role-based access for students & admins.



Room Management

Add/edit rooms with capacity, building, type and equipment.



Event Management

Create, edit and allocate rooms to events with expected turnout.



Registration Module

Students register for events; seats tracked against capacity.



Booking & Scheduling

Reserve rooms for time slots; detect overlapping bookings.



Occupancy Tracking

Log expected vs. actual attendance per event and room.



Conflict Alerts

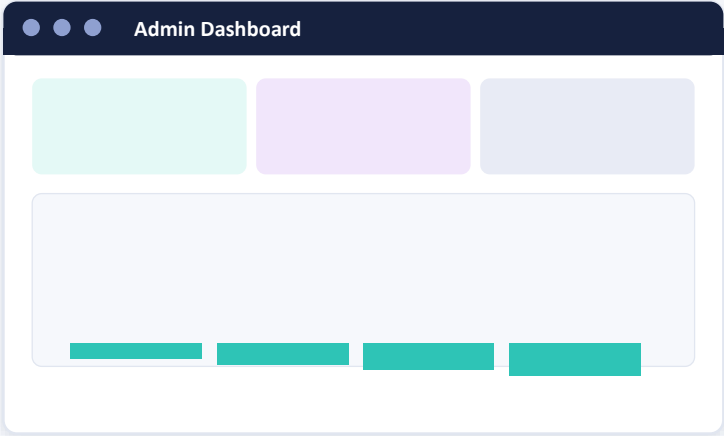
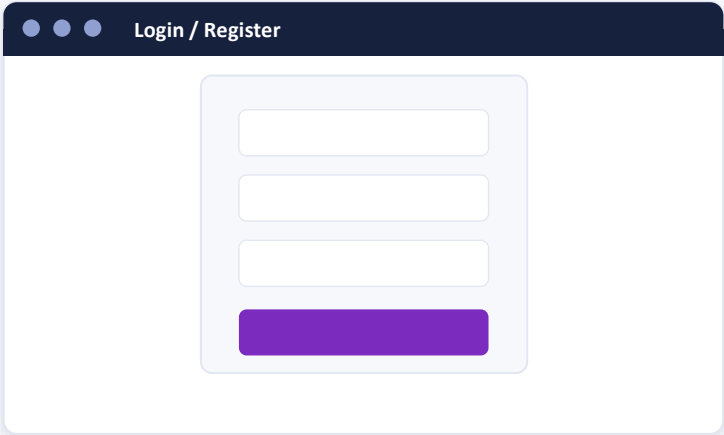
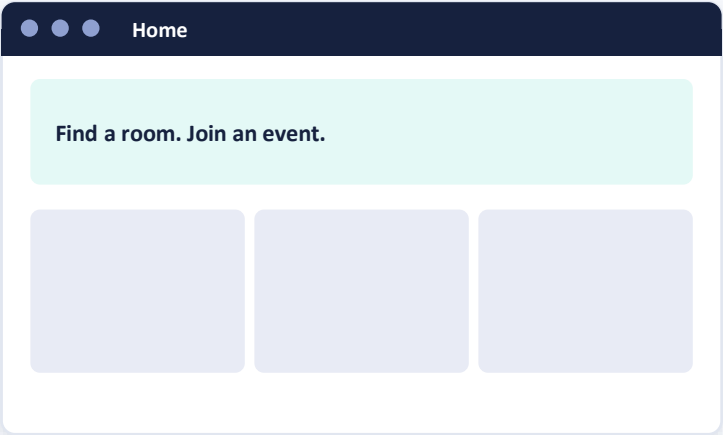
Flags double-bookings and capacity mismatches automatically.



Admin Dashboard

Central view of rooms, events, registrations and analytics.

Frontend Design



Home Page

Landing view surfacing quick room search and featured events.

Login / Registration

Role-aware auth screen for students and administrators.

Forms

Guided event registration capturing attendee details.

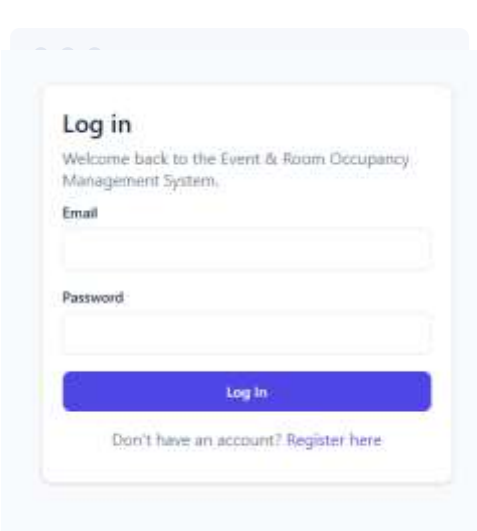
Dashboard

Admin control center for rooms, events and occupancy.

Frontend Implementation

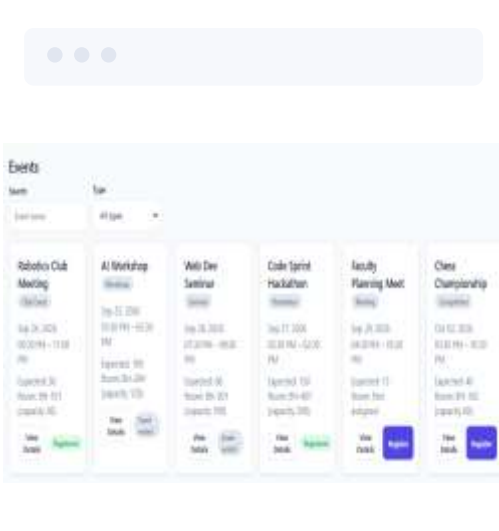
Working screens are being implemented in React and connected to the REST API. Frames below mark each page's current build status.





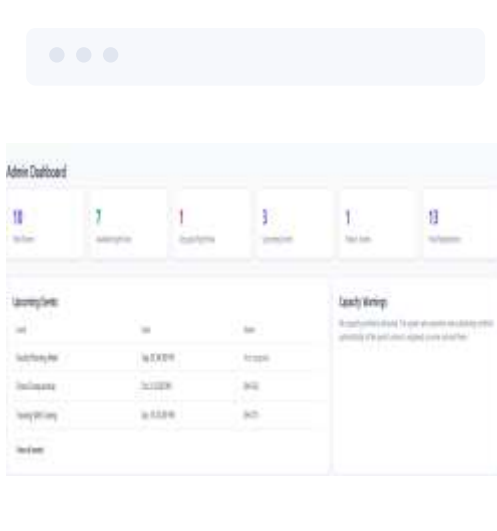
Login / Registration

In progress



Event Forms

Planned

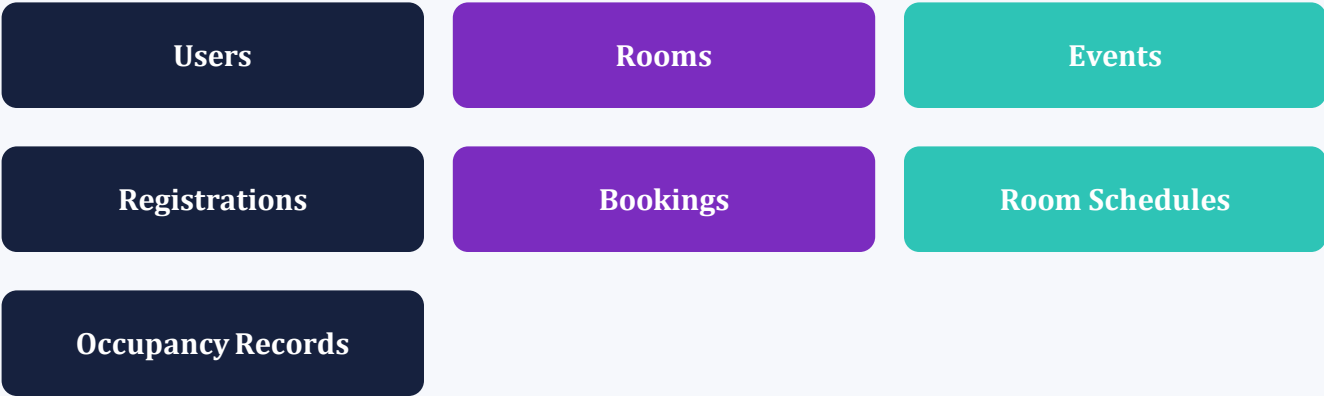


Admin Dashboard

Planned

Database Design

Seven related tables model the domain — normalized to 3NF so every non-key attribute depends only on its table's primary key.



KEY RELATIONSHIPS

- One User → Many Registrations
- One Event → Many Registrations
- One Event → One or More Bookings
- One Room → Many Room_Schedules
- One Event → One Room (via Bookings)

NORMALIZATION — 3NF

1NF

All attributes hold atomic, indivisible values.

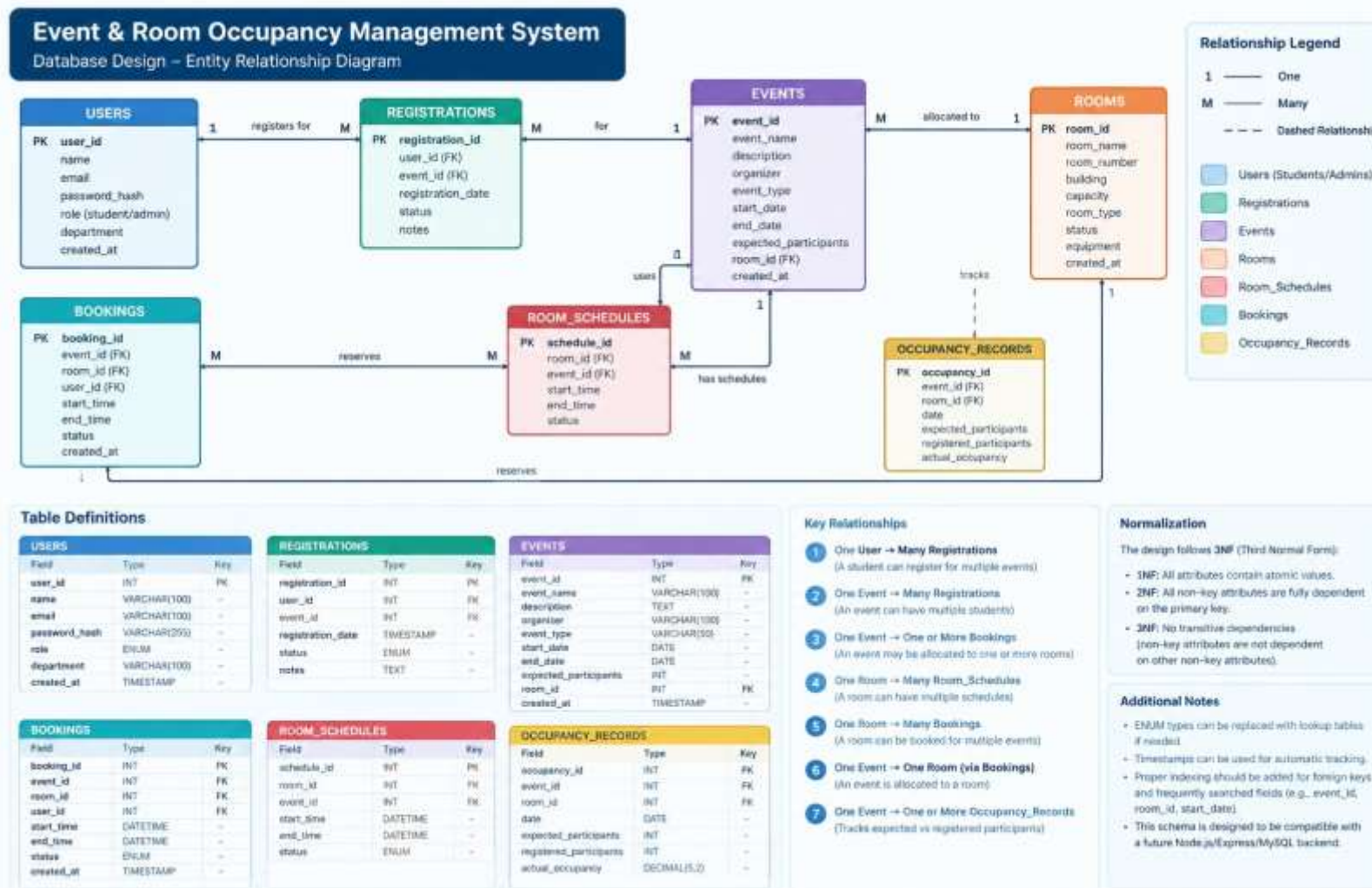
2NF

Every non-key attribute fully depends on the primary key.

3NF

No transitive dependencies between non-key attributes.

ER Diagram



Database Tables / Schema

USERS		
user_id	INT	PK
name	VARCHAR(100)	
email	VARCHAR(100)	
password_hash	VARCHAR(255)	
role	ENUM	
department	VARCHAR(100)	
created_at	TIMESTAMP	

EVENTS		
event_id	INT	PK
event_name	VARCHAR(100)	
description	TEXT	
organizer	VARCHAR(100)	
event_type	VARCHAR(50)	
start_date / end_date	DATE	
expected_participants	INT	
room_id	INT	FK

REGISTRATIONS		
registration_id	INT	PK
user_id	INT	FK
event_id	INT	FK
registration_date	TIMESTAMP	
status	ENUM	
notes	TEXT	

BOOKINGS		
booking_id	INT	PK
event_id / room_id	INT	FK
user_id	INT	FK
start_time / end_time	DATETIME	
status	ENUM	

ROOM_SCHEDULES		
schedule_id	INT	PK
room_id / event_id	INT	FK
start_time / end_time	DATETIME	
status	ENUM	

OCCUPANCY_RECORDS		
occupancy_id	INT	PK
event_id / room_id	INT	FK
date	DATE	
expected_participants	INT	
registered_participants	INT	
actual_occupancy	DECIMAL(5,2)	

ILLUSTRATIVE DATA

Sample Database Records

Representative rows showing how EVENTS and ROOMS data will be stored (sample values for demonstration).

EVENTS

event_id	event_name	event_type	start_date	expected_participants	room_id
101	Code Sprint Hackathon	Hackathon	2026-09-20	120	4
102	AI & ML Workshop	Workshop	2026-09-25	60	2
103	Cultural Fest Meet	Seminar	2026-10-02	200	6

ROOMS

room_id	room_name	building	capacity	room_type	status
2	Seminar Hall B	Block B	80	Seminar Hall	Available
4	Innovation Lab	Block C	150	Lab	Reserved
6	Main Auditorium	Block A	250	Auditorium	Available

Current Progress

COMPLETED

- ✓ Problem statement, objectives & scope finalized
- ✓ Proposed system & architecture designed
- ✓ ER diagram & normalized (3NF) schema completed
- ✓ Database tables created in MySQL Workbench
- ✓ Frontend wireframes for key pages designed
- ✓ Tech stack finalized: React, Node, Express, MySQL

IN PROGRESS

- ① React frontend build-out (home, login, forms, dashboard)
- ① Express REST API endpoints for rooms & events
- ① Booking conflict-detection logic
- ① Connecting frontend forms to backend APIs
- ① Seeding sample data for demo walkthroughs
- ① Postman test collections for API validation

Challenges & Future Work

CHALLENGES FACED

Concurrent bookings

Preventing race conditions when two users book the same room at once.

Conflict-detection logic

Designing reliable overlap checks across rooms, dates and time ranges.

Occupancy estimation

Capturing accurate real-time headcounts versus registered numbers.

FUTURE WORK

Notifications

Email/SMS alerts for registration confirmation & schedule changes.

Analytics dashboard

Trend reports on room utilization and event turnout over time.

Mobile-friendly PWA

Extend the responsive web app for on-the-go access.

Conclusion

The Event & Room Occupancy Management System now has a finalized database design, a normalized 7-table schema, and wireframed React screens. With the REST API and conflict-detection logic underway, the project is on track to deliver a centralized platform that gives students real-time room visibility and equips administrators with the tools to plan, allocate and monitor campus spaces confidently.

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Database Tables

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Core Modules

3NF

Normalized

React+Node

Stack

Thank you — questions welcome.